Università degli studi di Padova

Stochastic Methods for Finance

Report 7: Pricing and Sensitivity Analysis of Barrier Options

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1 Introduction

This report details the implementation and analysis of Down-and-Out (DO) and Down-and-In (DI) European Call options. We evaluate these instruments using closed-form analytical solutions, Monte Carlo simulations, and numerical methods for implied volatility extraction.

2 Analytical Pricing and Model Framework

The valuation of barrier options in this study is based on the closed-form solutions derived from the Black-Scholes framework, specifically utilizing the reflection principle for Geometric Brownian Motion. We implement the pricing formulas for European Down-and-Out (DO) and Down-and-In (DI) call options as defined in Propositions 18.17 and 18.24 of Björk's book.

We fixed parameters as following:

Parameter	Symbol	Value
Initial Spot Price	x	150
Barrier Level	B	130
Risk-free Interest Rate	r	0.02
Volatility	σ	0.40
Time to Maturity	T-t	1 Year
Strike Price (Case 1: $K < B$)	K_1	120
Strike Price (Case 2: $K > B$)	K_2	140

2.1 Down-and-Out European Call

According to Proposition 18.17, the price of a down-and-out European call option $\mathbf{C}_{LO}(t, s; K)$ depends on the relationship between the barrier L and the strike price K.

For $L < K$:

$$\mathbf{C}_{LO}(t, s; K) = \mathbf{C}(t, s; K) - \left(\frac{L}{s}\right)^{\frac{2\tilde{r}}{\sigma^2}} \mathbf{C}\left(t, \frac{L^2}{s}; K\right) \tag{1}$$

For $L > K$:

$$\begin{aligned}\mathbf{C}_{LO}(t, s; K) =& \mathbf{C}(t, s; K) + (L - K)\mathbf{H}(t, s; L) \\ &- \left(\frac{L}{s}\right)^{\frac{2\tilde{r}}{\sigma^2}} \left\{\mathbf{C}\left(t, \frac{L^2}{s}; K\right) + (L - K)\mathbf{H}\left(t, \frac{L^2}{s}; L\right)\right\}\end{aligned} \tag{2}$$

Where $\mathbf{C}$ denotes the standard European call price, $\mathbf{H}$ denotes the cash-or-nothing digital call price, and $\tilde{r} = r - \frac{1}{2}\sigma^2$ represents the drift of the log-process.

```
Case K1<B       DO: 6.561305727987502   Vanilla: 29.86934486389501
Case K2<B       DO: 4.725234990774199   Vanilla: 20.480611503003495
```

2.2 Down-and-In European Call

Proposition 18.24 provides the complementary pricing for the down-and-in European call $\mathbf{C}_{LI}(t, s; K)$.

For $L < K$:

$$\mathbf{C}_{LI}(t, s; K) = \left(\frac{L}{s}\right)^{\frac{2\tilde{r}}{\sigma^2}} \mathbf{C}\left(t, \frac{L^2}{s}; K\right) \tag{3}$$

For $L > K$:

$$\begin{aligned}\mathbf{C}_{LI}(t, s; K) = \left(\frac{L}{s}\right)^{\frac{2\tilde{r}}{\sigma^2}} \left\{\mathbf{C}\left(t, \frac{L^2}{s}; K\right) + (L-K)\mathbf{H}\left(t, \frac{L^2}{s}; L\right)\right\} \\ - (L-K)\mathbf{H}(t, s; L)\end{aligned} \tag{4}$$

```
Case K1<B       DI:  23.308039135907507         Vanilla:  29.86934486389501
Case K2>B       DI:  15.755376512229295         Vanilla:  20.480611503003495
```

2.3 In-Out Parity Verification

A fundamental property of these barrier options is the In-Out Parity, which states that the sum of the "in" and "out" components must recover the vanilla European call price:

$$\mathbf{C}_{Vanilla} = \mathbf{C}_{LI} + \mathbf{C}_{LO} \tag{5}$$

Our implementation confirms this identity with zero numerical error, validating the consistency of the reflected terms and the digital call adjustments ($\mathbf{H}$) used when the barrier exceeds the strike.

3 Sensitivity to the Barrier value

The barrier level L defines the survival probability of the option. By varying L from $0.5S_0$ to $0.99S_0$, we observe the transition from a "quasi-vanilla" instrument to a highly sensitive barrier instrument.

3.1 Price Behavior and Visual Convergence

As shown in the plots, the Down-and-Out price $V(L)$ is strictly decreasing with respect to the barrier level:

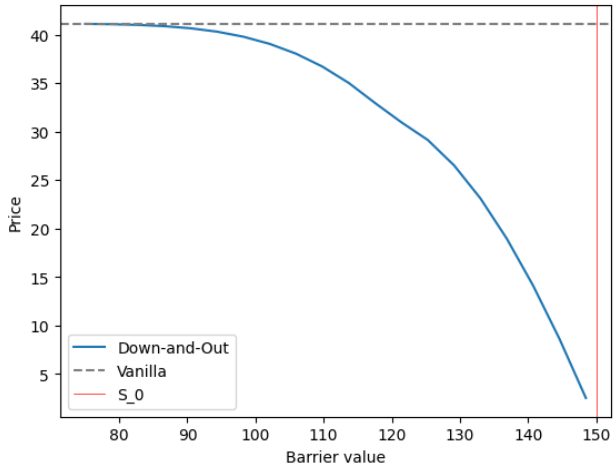


Figure 1: Price of Down-and-Out at varying L from $0.5S_0$ to $0.99S_0$

As $L \to 0$: The knockout event becomes improbable, and the price converges to the Vanilla price.

As $L \to S_0$: The probability of hitting the barrier approaches unity, causing the option value to vanish.

3.2 Numerical Verification of Limiting Regimes

Log-log regression analysis was employed to quantify convergence rates in both asymptotic regimes.

3.2.1 Convergence to Vanilla Price ($L \to 0$)

We analyzed the error $\epsilon_1 = |C^{DO}(L) - C^{vanilla}|$, observing a rate of $\alpha_1 \approx 16.36$. This high value is a numerical artifact; at low barrier levels, the knockout term $(L/S)^\eta$ falls below machine precision (10^{-16}), rendering the prices numerically identical.

3.2.2 Convergence to Zero ($L \to S_0$)

The error $\epsilon_2 = |C^{DO}(L) - 0|$ yielded a rate of $\alpha_2 \approx 0.91$. This near-linear behavior (≈ 1.0) confirms the price vanishes proportionally to the distance $(S_0 - L)$, consistent with the survival probability of Brownian motion near a boundary.

4 Delta Analysis and Barrier Risk

This section evaluates the directional sensitivity of the Down-and-Out call, comparing its risk profile to that of a standard vanilla European call, with a specific focus on the boundary behavior near L.

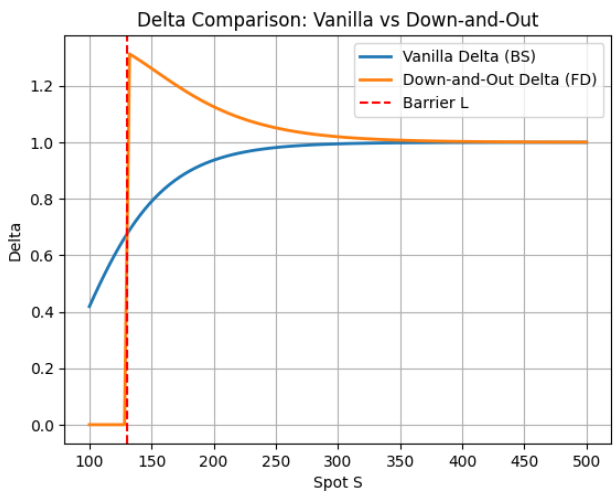
4.1 Numerical and Analytical Methodology

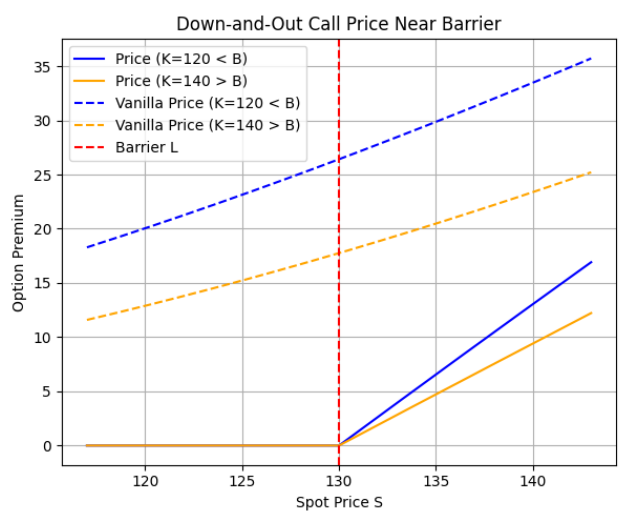
We compute the Delta (Δ) of the barrier option using a central finite difference approximation:

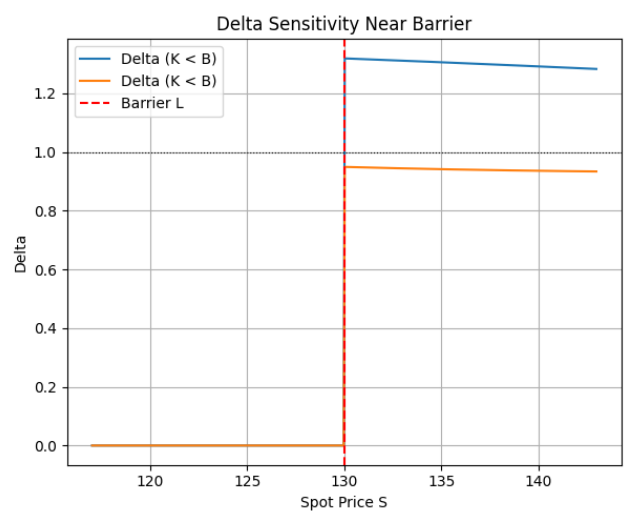
$$\Delta_{DO} \approx \frac{V(S_0 + \epsilon) - V(S_0 - \epsilon)}{2\epsilon} \tag{6}$$

where $V(S)$ is the closed-form price from Proposition 18.17. This numerical result is benchmarked against the analytical Delta of a vanilla European call, derived from the Black-Scholes formula:

$$\Delta_{Vanilla} = e^{-q\tau}\Phi(d_1) \tag{7}$$







4.2 Price Behavior and Comparison Near the Barrier

As the spot price S approaches the barrier L from above, the Down-and-Out (DO) call price converges to zero regardless of its moneyness. The transition from a vanilla to a barrier instrument introduces a radical shift in sensitivity behavior near the boundary.

- The Barrier Discount: Comparison between Down-and-Out and Vanilla instruments illustrates that the barrier option is strictly cheaper. However, this discount creates a sharp "elbow" in the price curve at $S = L$.
- Vanilla Delta: Exhibits a smooth, monotonic transition from 0 to 1, representing the probability of expiring in-the-money.
- Barrier Delta: For $S < L$, Delta is identically zero as the option is knocked out. For $S > L$, Delta is non-monotonic and exhibits a sharp "spike" near the boundary.

4.3 Analysis of Sensitivity and Delta Overshoot

The geometric "sharpness" of the price curve at the barrier translates into a significant discontinuity in the numerical Delta.

- Delta Overshoot: In cases where $K < L$ (reverse knock-out), Delta may jump above 1.0. This occurs because the option must shed its entire remaining intrinsic and time value instantaneously upon hitting the barrier.
- Negative Gamma: The rapid change in Delta as S nears L indicates extreme negative Gamma. Near the barrier, the option behaves like a leveraged position with extreme sensitivity to small movements.

4.4 Financial and Hedging Implications

The "cliff-edge" nature of these instruments creates high-stakes environments for delta-hedgers:

- Pinning Risk: As the spot approaches the barrier, a delta-neutral trader may need to maintain a long position exceeding 100% of the option's notional.
- Liquidation and Slippage: Upon touching the barrier, the option disappears. The entire hedge must be liquidated instantly, often leading to significant slippage in a falling market.
- Gap Risk: If the market gaps below the barrier, the option is knocked out while the hedge remains, leaving an unhedged long position in a declining market.

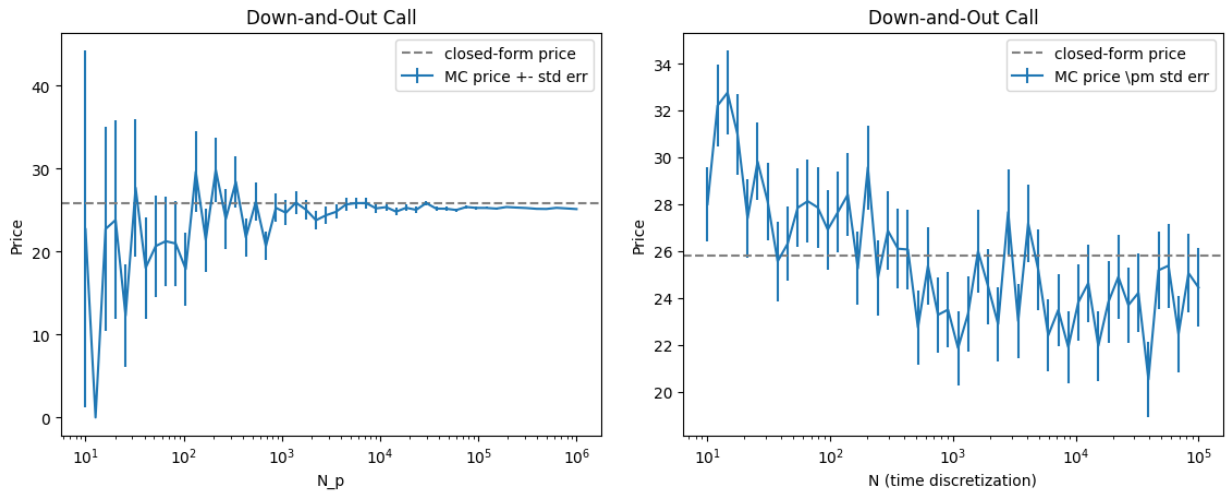
5 Monte Carlo Simulation and Discretization Bias

The Down-and-Out call price is estimated as the discounted expected payoff under the risk-neutral measure:

$$V_t = e^{-r(T-t)} E[(S_T - K)^+ \mathbf{1}_{S_u \geq B, \forall u \in (t,T)} | \mathcal{F}_t]$$

Trajectories were generated using the Euler-Maruyama discretization of the GBM SDE: $dS_t = rS_t dt + \sigma S_t dW_t$. The result is compared to the closed-form price.

```
MC results:  24.463465438165453 +- 1.6574880826014589
Closed-form price:  25.822021075388164
```



5.1 Statistical Convergence (N_p)

The estimator's precision is governed by the number of paths N_p. As N_p increases, the sample mean stabilizes and the standard error ($SE = \sigma_{payoff}/\sqrt{N_p}$) shrinks at the theoretical rate of $\mathcal{O}(1/\sqrt{N_p})$. At low N_p ($< 10^2$), the price exhibits high volatility due to outlier paths; however, by $N_p = 10^4$, the statistical noise is sufficiently dissipated.

5.2 Discretization Bias (N)

While N_p reduces noise, the number of time steps N determines the **discretization bias**.

- **Monitoring Gap:** At low N, the simulation consistently overestimates the price. This occurs because the barrier is only checked at discrete nodes, missing "intra-step" breaches where the price touches B and recovers between observations.
- **Convergence:** As N increases toward 10^5, monitoring frequency approaches continuity, and the MC price converges to the analytical benchmark.

6 Implied Volatility Extraction

To estimate the implied volatility (σ_{IV}), we solve for the root of the function $f(\sigma) = V_{model}(\sigma) - P_{mkt} = 0$. Given the potential non-monotonicity of barrier option prices with respect to volatility (due to knockout risk), the Bisection Method was implemented to ensure global convergence.

6.1 Numerical Implementation and Convergence

The bisection algorithm was configured with a search interval of $[0.001, 2.0]$ and a tolerance of 10^{-8}. As shown in the results, the algorithm successfully recovered the true volatility ($\sigma = 0.10$ and $\sigma = 0.30$) for the Down-and-Out model within approximately 30–32 iterations. This linear convergence confirms the robustness of the method.

6.2 Comparison with Vanilla Implied Volatility

We analyzed two distinct regimes by comparing the Barrier IV with the IV obtained by forcing the same market price into a standard Black-Scholes Vanilla model.

1. **Low Volatility Regime ($\sigma = 0.10, S = 150$):** The Vanilla IV (≈ 0.358) is significantly higher than the true Barrier IV (0.10). Here, the Vanilla model must overstate the volatility to account for the price suppression caused by the barrier.

```
Recovered Implied Vol of DO: 0.100000 in 32 steps
Recovered Implied Vol of Vanilla: 0.358372 in 31 steps
At sigma=0.1, DO Price: 39.36600901736014
At sigma=0.358372, Vanilla Price: 39.366002102172075
```

2. **Proximity Regime ($\sigma = 0.30, S = 135$):** As the spot price approaches the barrier, the Down-and-Out price drops below the Vanilla option's intrinsic value. In this case, the Vanilla solver fails to converge (hitting the upper bound of 2.0), as no standard volatility can reconcile a price lower than the European lower arbitrage bound: $C_{vanilla} \geq \max(S_t - Ke^{-r\tau}, 0)$.

```
Recovered Implied Vol of DO: 0.300000 in 30 steps
Recovered Implied Vol of Vanilla: 2.000000 in 500 steps
At sigma=0.3, DO Price: 7.228215683790104
At sigma=0.1, Vanilla Price: 17.860826943962863
```

7 Conclusion

Barrier options offer lower premiums at the cost of discontinuous risk profiles. This study highlights that while analytical formulas provide precision, the financial reality near the barrier involves high Gamma risk and model sensitivity that standard vanilla frameworks cannot capture.